Word Frequency in Poetry: Computational Insights into Groot Verseboek and the Formation of the Afrikaans Literary Canon

Smit, Mathilda
smitm1@ufs.ac.za
University of the Free State, South Africa

Strauss, Trudie
strausst@ufs.ac.za
University of the Free State, South Africa

This study explores the intersection of Digital Humanities and literary analysis through a statistical word frequency analysis of Groot Verseboek (Brink, 2008), a canonized anthology of Afrikaans poetry spanning three chronologically organized volumes. Groot Verseboek is compiled by André P Brink and poems included in this anthology can be regarded as part of the literary canon of Afrikaans. The concept of a canon involves a process of selection and validation, often reflecting dominant ideological, linguistic, and cultural norms. The anthologies' inclusion and/or exclusion of poets and poems solidifies certain linguistic, cultural, and thematic values as representative of the literary heritage in Afrikaans. This has profound implications for how the canon is understood and how Afrikaans literature is preserved, taught, and remembered.

Using accessible, language-agnostic computational methods – specifically basic word frequency analysis – this study identifies recurring content words and explores their potential to signal dominant themes and stylistic trends within this closed corpus. While the methodology is deliberately limited to frequency-based techniques due to current constraints in Afrikaans NLP tools, this approach provides a solid foundation that can be extended to more intricate computational analyses in future research.

The analysis is grounded in a broader interpretive framework that integrates cultural context and knowledge of literary movements, specifically in Afrikaans. Key questions include the extent to which word frequency mirrors shifts in socio-historical contexts and how the anthology reflects the politics of inclusion and exclusion. Ultimately, this interdisciplinary approach demonstrates how statistical insights, and qualitative literary analysis can complement one another, offering new perspectives on canon formation, thematic evolution, and cultural representation in Afrikaans literature.

Acknowledging methodologies from digital humanities approaches to text analysis (Arnold et al., 2019), and a collection of automatic tools particularly focussed on poetry (De Sisto et al., 2024), we note that these are not available for the Afrikaans language. The current study therefore starts with applying basic word frequency analysis to identify dominant themes and examines how these themes reflect the socio-historical contexts of different periods in Afrikaans poetry. This study also investigates shifts in lexical fields (i.e. themes), considering how the frequency and use of specific thematic vocabulary mirror broader changes within Afrikaans-speaking communities and the broader South African socio-political landscape. Kuipers' (2003) theory of the anthology/corpus dynamic informs the analysis of how power dynamics shape the selection and valorisation of themes, revealing the influence of dominant cultural and linguistic ideologies. While this research limits itself to foundational methods, it demonstrates how even basic computational tools, when combined with qualitative literary interpretation, can yield valuable insights into canon formation, thematic evolution, and cultural representation in Afrikaans literature.

Bibliography

Arnold, Taylor / Ballier, Nicolas / Lissón, Paula / Tilton, Laura (2019): "Beyond lexical frequencies: Using R for text analysis in the digital humanities", in: *Language Resources and Evaluation* 53, 4: 707 – 733 < https://www.jstor.org/stable/45286325 > [30.04.2025]

Brink, André P. (Ed.). (2008): *Groot verseboek* . Cape Town: Tafelberg.

De Sisto, Mirella / Hernández-Lorenzo, Laura / De la Rosa, Javier / Ros, Salvador / & González-Blanco, Elena (2024): "Understanding poetry using natural language processing tools: A survey", in: *Digital Scholarship in the Humanities* 39, 2: 500 – 521 < https://academic.oup.com/dsh/article/39/2/500/7602425 > [30.04.2025]

Green, Clarence (2017): "Introducing the Corpus of the Canon of Western Literature: A corpus for culturomics and stylistics" in: *Language and Literature* 26, 4: 282 – 299 < https://journals.sagepub.com/doi/10.1177/0963947017718996 > [20.09.2024]

Kuipers, Christopher M. (2003): “The anthology/corpus dynamic: A field theory of the canon” in: *College Literature* 30, 2: 51 – 71 < https://www.jstor.org/stable/25112719 > [20.09.2024]

R Core Team (2024): R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. < https://www.R-project.org >